

Cybersecurity Training Diary

Day 7: Encryption & Cryptography Basics

1. Topics Covered

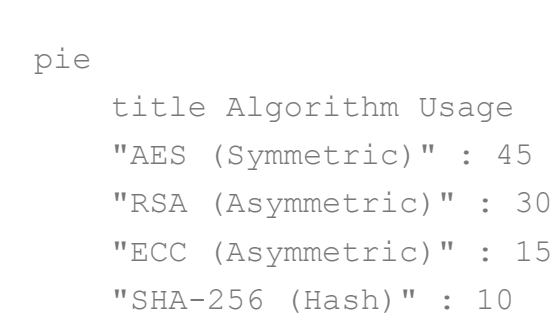
✓ Core Cryptographic Concepts

Term	Definition	Real-World Example
Plaintext	Original readable data	"Hello World"
Ciphertext	Encrypted data	"Khoor Zruog" (Caesar Cipher)
Key	Secret for encryption/decryption	Password, RSA private key

✓ Encryption Types

Type	How It Works	Use Case
Symmetric	Same key encrypts/decrypts (AES)	Full-disk encryption
Asymmetric	Public/private key pairs (RSA)	SSL/TLS, Bitcoin
Hashing	One-way transformation (SHA-256)	Password storage

✓ Common Algorithms



✓ Real-World Applications

- **SSL/TLS:** Padlock icon in browsers

- **Signal/WhatsApp:** End-to-end encryption
- **Blockchain:** Cryptographic hashing for Bitcoin

2. Key Takeaways

Symmetric = Fast, but key distribution is hard

Asymmetric = Secure, but computationally expensive

Hashes are irreversible (but rainbow tables can crack weak ones)

3. Practical Exercises

Exercise 1: Encrypt a File with OpenSSL

```
# AES-256 Encryption
openssl enc -aes-256-cbc -salt -in secret.txt -out secret.enc

# Decryption
openssl enc -d -aes-256-cbc -in secret.enc -out plaintext.txt
```

Exercise 2: Create Hashes

```
import hashlib
print(hashlib.sha256(b"password123").hexdigest())
# Output: ef92b778bafef771e89245b89ecbc08a44a4e166c06659911881f383d4473e94
```

4. Questions for Reflection

Why can't we use hashing for encryption?

What happens if you lose your private key in asymmetric crypto?

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